AAKASH K

Electronics and Communication Engineering
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PROFILE

Embedded Systems Engineer with a solid foundation in designing, developing, and debugging firmware for resource-constrained devices. Proficient in C/C++, RTOS (FreeRTOS/Zephyr), and microcontroller platforms such as ARM Cortex-M, STM32, and ESP32. Skilled in peripheral interfaces (I2C, SPI, UART, CAN), low-power design, and hardware bring-up, with a strong foundation in delivering reliable, production-ready products from prototype to release.

PROJECTS

SPOTFINDER IoT – Real-Time Smart Parking System

- Developed a real-time smart parking system using STM32F103C8T6 and IR sensors for vehicle detection.
- Integrated ESP8266, Wi-Fi, MQTT, and JSON for cloud-based real-time parking data communication.
- Designed a scalable IoT architecture with mobile/web access to provide instant parking availability.

Occlusion-Aware Dynamic Camera Ranking for Multi-Floor Smart Building Surveillance

- Developed a Dynamic Camera Ranking Algorithm for multi-floor smart building surveillance under time-varying priorities.
- Implemented Occlusion-Aware Camera Selection with Priority-Based Ranking to optimize surveillance coverage.
- Incorporated Battery/Energy Constraints for efficient camera operation and resource-aware monitoring.

EDUCATION

B.E. Electronics and Communication Engineering | CGPA: 7.24 (up to 6th Sem)
K.S.R. College of Engineering

12th (SSLC) | Saraswati Matric Higher Secondary School | 2022 – 2023 | 78%
10th (SSC) | Saraswati Matric Higher Secondary School | 2020 – 2021 | 100%

CERTIFICATIONS

- Taras Systems & Solutions Level 1 – Advanced Embedded Systems & IoT: STM32 programming, sensor interfacing, I2C/SPI/UART/CAN protocols.
- Taras Systems & Solutions Level 2 – IoT & Edge AI Cloud integration, data communication, Python for edge inference.
- Chip Crafts – Advanced PCB Design: PCB layout, circuit design rules, and EDA tools for hardware implementation.
- C/C++ Programming (Certified) | MS Office (Certified)

TECHNICAL SKILLS

Languages: C, Embedded C, C++

Microcontrollers: STM32F103C8T6 (ARM Cortex-M3), ESP32, ESP8266, Arduino UNO (AVR)

Architecture: ARM Cortex-M 32-bit MCU programming, memory-mapped I/O, NVIC interrupt handling

IDEs & Tools: STM32CubeIDE, Arduino IDE, Proteus (Simulation & Schematic), Git

Lab & Debug: Oscilloscope, Multimeter, DC Power Supply, Logic-level debugging, H/W-S/W integration testing

Protocols: I2C, SPI, UART/USART, CAN

RTOS & Cloud: FreeRTOS (fundamentals), Blynk IoT Cloud, MQTT protocol

Soft Skills: Teamwork, Adaptability, Time Management, Communication, Problem Solving